

Medvale Internal Medicine

Chapter 12B: Ambulatory Respiratory, ENT, Eye, GI/GU, Sleep, Fatigue, and Common Clinic Complaints

Outpatient triage, antibiotic stewardship, symptom algorithms, red flags, and management frameworks

Chapter purpose

This chapter finishes the common-clinic portion of ambulatory internal medicine. The focus is not to replace organ-system chapters, but to teach how common outpatient complaints are triaged, how dangerous disease is excluded, when antibiotics or imaging are appropriate, and how follow-up is structured. The chapter emphasizes red flags, diagnostic stewardship, watchful waiting, return precautions, and treatment thresholds that are commonly tested and commonly used in clinic.

1. Ambulatory triage framework for common clinic complaints

Ambulatory medicine is built around probability and safety. Most outpatient visits involve self-limited illness or chronic disease follow-up, but the clinician must rapidly identify the minority of patients whose complaint represents sepsis, respiratory failure, threatened vision, CNS infection, acute abdomen, pyelonephritis, ectopic pregnancy, malignancy, medication toxicity, or decompensated cardiopulmonary disease. The practical structure is: stabilize first, decide site of care, identify red flags, select a narrow diagnostic test set, treat symptomatically when appropriate, and create explicit return precautions.

For Graph-RAG extraction, every ambulatory complaint can be organized into four linked nodes: the symptom pattern, the red-flag pattern, the diagnostic test threshold, and the treatment/follow-up threshold. For example, acute cough without abnormal vital signs is usually viral bronchitis; acute cough with fever, tachycardia, hypoxemia, focal lung findings, or frailty shifts toward pneumonia evaluation. Similarly, red eye with itch and bilateral watery discharge is usually conjunctivitis, whereas red eye with pain, photophobia, decreased vision, corneal opacity, or contact lens use requires urgent ophthalmologic evaluation.

Clinic decision point	Low-risk outpatient pattern	Escalation pattern	Immediate action
Vital signs	Normal BP, HR, RR, O2 saturation, temperature; patient appears well	Hypotension, hypoxemia, persistent tachycardia, tachypnea, fever with toxicity	ED transfer or urgent same-day evaluation; do not anchor on benign outpatient diagnosis
Time course	Mild symptoms with expected natural history and clear follow-up	Rapid progression, abrupt severe onset, recurrent visits without improvement	Reassess diagnosis, complications, medication effects, and need for imaging/labs
Organ threat	No neurologic deficit, no vision loss, no respiratory distress, no renal compromise	Focal neuro signs, eye pain/vision loss, dyspnea at rest, oliguria, chest pain	Emergency workup and specialty consultation as needed
Treatment decision	Symptomatic care, narrow testing, shared decision-making	Potential need for IV therapy, airway support, procedure, or monitored setting	Refer to ED/inpatient setting rather than continuing routine clinic management
Follow-up	Self-limited illness with expected improvement window	No reliable follow-up, immunocompromise, advanced age/frailty, pregnancy, high-risk comorbidity	Lower threshold for labs, imaging, empiric therapy, or referral

2. Acute cough, URI, acute bronchitis, and outpatient pneumonia triage

Upper respiratory infections and acute cough are among the most common outpatient complaints. The key ambulatory skill is separating viral illness and acute bronchitis from pneumonia, pertussis, asthma/COPD exacerbation, heart failure, pulmonary embolism, medication cough, and malignancy. In otherwise healthy adults, acute bronchitis is usually viral and cough can persist for several weeks even after systemic symptoms improve. Purulent or colored sputum alone does not prove bacterial infection, and antibiotics do not meaningfully shorten uncomplicated viral bronchitis. The clinician should instead evaluate vital signs, lung exam, duration, exposure history, comorbid disease, and complications.

Pneumonia becomes more likely when cough is accompanied by fever, tachycardia, tachypnea, hypoxemia, focal crackles, egophony, pleuritic pain, rigors, or toxic appearance. In older adults, pneumonia may present with confusion, weakness, anorexia, falls, or functional decline rather than dramatic cough. Outpatient treatment is reasonable only when the patient is clinically stable, able to take oral therapy, has adequate oxygenation, and can follow up. Hospital evaluation is appropriate for hypoxemia, sepsis physiology, inability to tolerate oral intake, severe comorbid illness, altered mental status, or high risk by clinical judgment.

Condition	Typical pattern	Testing threshold	Management
Viral URI	Rhinorrhea, congestion, sore throat, cough, low-grade fever, myalgias; symptoms peak early and improve gradually	No routine testing unless influenza/COVID/RSV result would change isolation or antiviral decisions	Supportive care: fluids, analgesics/antipyretics, saline, honey/dextromethorphan if appropriate; return for dyspnea, fever persistence, worsening after improvement
Acute bronchitis	Acute cough, often after URI; sputum may be clear or colored; normal oxygenation and no focal pneumonia signs	Chest radiograph if abnormal vital signs, focal lung exam, hypoxemia, older frail patient, or concern for pneumonia	No antibiotics for uncomplicated disease. Treat symptoms; consider bronchodilator only if wheeze/bronchospasm
Pertussis	Paroxysmal cough, post-tussive emesis, inspiratory whoop, known exposure, underimmunized contact, prolonged cough	PCR/culture depending on time course and public health guidance	Macrolide treatment mainly reduces transmission; isolate and treat close contacts when indicated
Outpatient pneumonia	Cough plus fever, pleuritic pain, dyspnea, focal crackles/egophony, abnormal vitals, or infiltrate	Chest radiography when pneumonia suspected; pulse oximetry strongly useful	Oral antibiotics guided by comorbidity and local resistance; reassess if no improvement in 48-72 h or worsening
Postinfectious cough	Dry cough lasting 3-8 weeks after URI; patient otherwise improving	Evaluate if hemoptysis, weight loss, abnormal exam, smoking risk, immunocompromise, or cough >8 weeks	Reassurance, avoid irritants, symptomatic therapy; consider asthma, GERD, upper airway cough syndrome if persistent

Antibiotic stewardship anchor

Antibiotics are not indicated for uncomplicated acute bronchitis when pneumonia is not suspected. The outpatient note should document why pneumonia is unlikely: normal oxygenation, no fever/toxicity, no focal lung findings, and absence of high-risk comorbid features. Patient satisfaction is improved by explaining the expected cough duration and giving clear return precautions rather than prescribing unnecessary antibiotics.

3. Sinusitis, pharyngitis, otitis, and ENT complaints

ENT complaints frequently drive unnecessary antibiotic use. Most acute rhinosinusitis is viral. Bacterial sinusitis is suggested by persistent symptoms without improvement for at least 10 days, severe onset with high fever and purulent nasal discharge or facial pain for several days, or a double-worsening pattern in which symptoms initially improve and then worsen. Sinus imaging is not needed for routine uncomplicated acute sinusitis; imaging is reserved for suspected orbital/intracranial complications, recurrent disease with surgical planning, immunocompromise, severe atypical presentation, or concern for alternative diagnosis.

Pharyngitis requires a different logic. Viral pharyngitis is suggested by cough, rhinorrhea, hoarseness, oral ulcers, conjunctivitis, and diffuse URI symptoms. Group A streptococcal pharyngitis is suggested by fever, tonsillar exudates, tender anterior cervical nodes, and absence of cough. Adults should not receive empiric antibiotics for sore throat unless testing or a high-probability local protocol supports it. Infectious mononucleosis should be considered with marked fatigue, posterior cervical lymphadenopathy, hepatosplenomegaly, or prolonged pharyngitis, especially in adolescents and young adults.

Complaint	Key clues	Do not miss	Treatment logic
Acute bacterial rhinosinusitis	Persistent ≥ 10 days, severe fever/purulence/facial pain, or double worsening	Orbital cellulitis, cavernous sinus thrombosis, meningitis, invasive fungal sinusitis in immunocompromised host	Supportive care for viral. If bacterial pattern, amoxicillin-clavulanate is typical first-line unless allergy/resistance risk
Pharyngitis	Strep: fever, exudates, tender anterior nodes, no cough. Viral: cough/rhinorrhea/hoarseness/oral ulcers	Peritonsillar abscess, epiglottitis, Lemierre syndrome, airway compromise, mono splenic rupture risk	Use rapid strep/NAAT/culture based on age/local pathway. Treat confirmed strep; supportive care for viral
Otitis media	Ear pain, fever, bulging erythematous immobile tympanic membrane; adults less common than children	Mastoiditis, malignant otitis externa, cholesteatoma, sudden hearing loss	Analgesia always. Antibiotics if clear bacterial AOM in adults; reassess persistent symptoms
Otitis externa	Ear canal pain, pruritus, discharge, pain with tragus/pinna movement; swimmer or eardrum/trauma history	Malignant otitis externa in diabetes/immunocompromise; cranial nerve deficits	Topical antibiotic +/- steroid drops; keep ear dry; avoid traumatic cleaning
Hoarseness	Usually viral laryngitis or voice overuse if acute	Airway symptoms, tobacco/alcohol risk, neck mass, hemoptysis, dysphagia, persistent >3-4 weeks	Supportive care if acute; laryngoscopy for persistent or red-flag hoarseness

ENT red flags requiring urgent evaluation

- Sore throat with drooling, stridor, tripod positioning, muffled hot-potato voice, trismus, uvular deviation, neck swelling, or inability to handle secretions.

- Sinus symptoms with proptosis, ophthalmoplegia, decreased vision, severe frontal headache, altered mental status, meningeal signs, or immunocompromise.
- Ear pain with diabetes, immunosuppression, granulation tissue in the canal, cranial nerve deficits, severe persistent otalgia, or mastoid tenderness.
- Sudden sensorineural hearing loss, vertigo with neurologic deficits, or persistent unilateral tinnitus with hearing loss.

4. Red eye, vision complaints, and eye triage

In primary care, red eye is common but potentially dangerous because benign conjunctivitis and sight-threatening keratitis, uveitis, acute angle-closure glaucoma, scleritis, and orbital cellulitis can initially look similar. The first outpatient step is to determine whether vision is threatened. Always ask about pain, photophobia, decreased visual acuity, contact lens use, trauma/foreign body, chemical exposure, headache/nausea/halos, immunosuppression, and vesicular facial rash. Visual acuity is a vital sign for the eye and should be documented before treatment or referral whenever possible.

Conjunctivitis usually causes diffuse conjunctival injection with discharge but preserved vision and minimal pain. Viral conjunctivitis often has watery discharge and preauricular adenopathy; allergic conjunctivitis is intensely itchy and bilateral; bacterial conjunctivitis has purulent discharge and eyelids stuck together, though clinical overlap is substantial. Keratitis is more concerning when pain, photophobia, foreign-body sensation, corneal opacity, or contact lens use is present. Uveitis tends to cause consensual photophobia, ciliary flush, aching pain, and blurred vision. Acute angle-closure glaucoma causes severe eye pain, headache, nausea/vomiting, halos, cloudy cornea, mid-dilated pupil, and high intraocular pressure.

Eye pattern	Likely diagnosis	Red-flag features	Disposition
Itchy, bilateral, watery/stringy discharge	Allergic conjunctivitis	Vision loss, severe pain, photophobia absent in routine allergy	Topical antihistamine/mast cell stabilizer, artificial tears, allergen avoidance
Watery discharge, URI exposure, gritty sensation	Viral conjunctivitis	Severe pain, photophobia, corneal opacity, contact lens use	Supportive care, hygiene; avoid routine antibiotics
Purulent discharge, eyelids stuck shut	Bacterial conjunctivitis	Contact lens use, corneal involvement, vision change	Topical antibiotics may shorten course; urgent eval if contact lens/corneal signs
Painful red eye with photophobia or corneal staining	Keratitis/corneal abrasion/ulcer	Contact lens use, corneal opacity, dendritic lesions, decreased vision	Same-day ophthalmology for keratitis/ulcer; fluorescein exam if available
Deep ache, ciliary flush, consensual photophobia	Anterior uveitis/iritis	Autoimmune disease, irregular pupil, decreased vision	Urgent ophthalmology; do not start steroid drops without eye specialist
Severe pain, headache, halos, nausea/vomiting, cloudy cornea	Acute angle-closure glaucoma	Mid-dilated fixed pupil, high IOP, systemic symptoms	Emergency ophthalmology/ED
Painless red patch, normal vision	Subconjunctival hemorrhage	Trauma, anticoagulation, recurrent episodes, abnormal BP/bleeding	Reassurance if isolated; check BP/coagulation context

Do-not-miss eye rule

Any red eye with moderate-to-severe pain, photophobia, decreased visual acuity, corneal opacity, abnormal pupil, contact lens use, trauma/chemical exposure, vesicular rash near the eye, or headache/nausea/halos is not routine conjunctivitis until proven otherwise.

5. Dyspepsia, GERD follow-up, nausea, constipation, and diarrhea

Outpatient gastrointestinal symptoms are often chronic, recurrent, and functional, but the ambulatory clinician must identify alarm features and decide when empiric therapy is safe. Dyspepsia refers to epigastric pain or discomfort, early satiety, postprandial fullness, bloating, or nausea. GERD refers to reflux of gastric contents causing heartburn, regurgitation, cough/hoarseness, or esophagitis. The first questions are age, weight loss, GI bleeding, anemia, progressive dysphagia, odynophagia, persistent vomiting, family history of upper GI malignancy, prior ulcer disease, NSAID use, anticoagulation, and systemic illness. In younger patients without alarm features, H. pylori test-and-treat or empiric acid suppression is usually reasonable; older patients or those with alarm features need endoscopic evaluation.

Constipation should be described by stool frequency, stool form, straining, incomplete evacuation, manual maneuvers, and duration rather than by frequency alone. Medication review is essential because opioids, anticholinergics, calcium channel blockers, iron, calcium, tricyclic antidepressants, antihistamines, antipsychotics, and diuretics are common contributors. Alarm signs include new constipation in an older adult, iron deficiency anemia, hematochezia, weight loss, obstructive symptoms, severe pain, family history of colorectal cancer, or abnormal rectal exam. Diarrhea should be separated into acute versus chronic, inflammatory versus watery, and infectious versus noninfectious.

GI complaint	Initial outpatient approach	Alarm features	Common management pathway
Dyspepsia	Medication review; NSAIDs/alcohol; H. pylori test-and-treat or PPI trial depending on age/risk	Weight loss, anemia, melena, hematemesis, progressive dysphagia, odynophagia, persistent vomiting, older age threshold	Endoscopy for alarm/high-risk; eradicate H. pylori if positive; PPI trial if negative or reflux-predominant
GERD	Lifestyle triggers, nocturnal symptoms, obesity, pregnancy, medications; trial PPI if typical symptoms	Dysphagia, odynophagia, GI bleeding, weight loss, anemia, recurrent vomiting	PPI 4-8 weeks then step down if controlled; endoscopy for alarm or refractory symptoms
Nausea/vomiting	Hydration status, pregnancy possibility, meds/toxins, infection, CNS/vestibular signs, obstruction symptoms	Severe dehydration, bilious emesis, peritonitis, neuro signs, chest pain, severe headache, pregnancy complications	Oral rehydration/antiemetic if benign; labs/imaging/ED for red flags
Constipation	Diet/fluid/activity, meds, stool form, rectal symptoms, hypothyroid/hypercalcemia risk	New onset older adult, blood, anemia, weight loss, severe pain, obstruction, abnormal rectal exam	Fiber/osmotic laxative, treat cause, colonoscopy if alarm or screening due
Acute diarrhea	Assess dehydration, blood/fever, exposure/travel/antibiotics, outbreak risk	Bloody diarrhea, high fever, severe pain, sepsis, immunocompromise, pregnancy, dehydration	Oral rehydration; stool studies for red flags; avoid antimotility agents in suspected invasive infection
Chronic diarrhea	Duration >4 weeks; categorize watery/fatty/inflammatory; review meds, diet, IBD/celiac/endocrine risk	Weight loss, nocturnal symptoms, blood, anemia, hypoalbuminemia, family history, abnormal labs	CBC/CMP/TSH/celiac testing/stool markers guided by pattern; colonoscopy if alarm/inflammatory

6. UTI, pyelonephritis, hematuria, vaginitis, and common GU complaints

Outpatient GU complaints require careful pattern recognition because urinary symptoms can represent cystitis, pyelonephritis, urethritis/STI, nephrolithiasis, vaginitis, prostatitis, pregnancy-related disease, or malignancy. Acute uncomplicated cystitis classically causes dysuria, frequency, urgency, and suprapubic discomfort without fever, flank pain, systemic toxicity, or vaginal symptoms. When dysuria is accompanied by vaginal discharge, irritation, pelvic pain, new sexual partner, or cervicitis signs, STI or vaginitis evaluation becomes more important than empiric cystitis treatment.

Pyelonephritis is suggested by fever, flank pain, costovertebral angle tenderness, nausea/vomiting, systemic symptoms, or sepsis physiology. These patients generally require urine culture, pregnancy consideration, assessment of hydration and oral tolerance, and an antibiotic strategy that reaches renal tissue. Men, pregnancy, anatomic urinary abnormalities, obstruction, stones, immunocompromise, renal transplant, recent instrumentation, recurrent resistant organisms, and severe comorbidity make UTI complicated and lower the threshold for culture, imaging, or ED referral.

GU pattern	Most likely diagnosis	Diagnostic move	Treatment / escalation
Dysuria + frequency/urgency, no vaginal symptoms, no fever/flank pain	Acute uncomplicated cystitis	UA may support; culture if recurrent, atypical, pregnancy, complicated risk	Nitrofurantoin 100 mg BID x5 days, TMP-SMX DS BID x3 days if local resistance acceptable, or fosfomycin 3 g once; adjust for renal function/allergy
Dysuria + vaginal discharge/irritation	Vaginitis, cervicitis, urethritis/STI	Pelvic exam or self-swab NAAT/wet mount/pH/whiff depending setting; GC/CT/trich testing	Treat based on syndrome and tests; avoid anchoring on UTI
Fever/flank pain/CVA tenderness +/- vomiting	Pyelonephritis	UA and urine culture; assess pregnancy, sepsis, obstruction, oral tolerance	Outpatient only if stable. ED/inpatient for sepsis, vomiting, pregnancy, obstruction, immunocompromise, frailty
Colicky flank pain radiating to groin + hematuria	Nephrolithiasis	UA, pregnancy test if applicable, renal function; imaging based on risk/severity	Analgesia, hydration; urgent if infected obstruction, AKI, solitary kidney, uncontrolled pain/vomiting
Gross hematuria or persistent microscopic hematuria	Stone, infection, malignancy, glomerular disease, anticoagulation-related uncovering pathology	Repeat UA after infection treatment; evaluate protein/casts; risk-stratify for urology	Do not dismiss gross hematuria. Refer for cystoscopy/imaging based on age/risk
Male pelvic/perineal pain, obstructive urinary symptoms, fever	Prostatitis	UA/culture; rectal exam gently if needed; avoid vigorous prostate massage	Antibiotics with prostate penetration; ED if sepsis/retention

Vaginitis pattern recognition

Condition	Typical discharge / symptoms	pH / microscopy clues	Treatment concept
Bacterial vaginosis	Thin gray/white discharge, fishy odor, minimal inflammation	pH >4.5, clue cells, positive whiff	Metronidazole or clindamycin regimen; evaluate STI risk

Condition	Typical discharge / symptoms	pH / microscopy clues	Treatment concept
Vulvovaginal candidiasis	Thick white discharge, vulvar pruritus/burning, erythema	Usually normal pH; budding yeast/pseudohyphae	Topical azole or oral fluconazole if appropriate; complicated/recurrent needs longer strategy
Trichomoniasis	Frothy yellow-green discharge, irritation, dysuria, cervicitis	pH >4.5; motile trichomonads if seen; NAAT more sensitive	Metronidazole/tinidazole; treat partners and test for other STIs
Atrophic/genitourinary syndrome of menopause	Dryness, dyspareunia, recurrent irritation/UTI-like symptoms	Elevated pH, pale friable mucosa	Vaginal moisturizers/lubricants; topical estrogen when appropriate

7. Fatigue, sleep complaints, and obstructive sleep apnea

Fatigue is one of the highest-yield ambulatory symptoms because it is common, nonspecific, and strongly affected by sleep, mood, medications, endocrine disease, anemia, infection, malignancy, cardiopulmonary disease, and social stressors. The outpatient goal is not to order every test; it is to decide whether fatigue is physiologic/situational, psychiatric/sleep-related, medication-related, endocrine/metabolic, inflammatory/infectious, cardiopulmonary, neurologic, or malignant. Time course matters: acute fatigue after viral illness differs from progressive fatigue with weight loss and night sweats. Functional impairment and objective weakness should be separated from subjective tiredness.

Initial laboratory testing is often reasonable when fatigue is persistent, unexplained, or associated with systemic symptoms. A practical first set includes CBC, CMP, TSH, and targeted testing based on history such as pregnancy test, ferritin/iron studies, A1c, ESR/CRP, HIV/hepatitis testing, CK, B12, urinalysis, or sleep testing. A normal basic evaluation does not mean symptoms are not real; it means the next step should emphasize sleep, mood, substance use, medication effects, activity pacing, and follow-up rather than repetitive low-yield testing.

Fatigue clue	Potential diagnostic direction	High-yield next step
Snoring, witnessed apneas, morning headache, daytime sleepiness, resistant HTN, obesity/large neck	Obstructive sleep apnea	STOP-BANG screen; home sleep apnea testing or polysomnography based on risk/comorbidity
Anhedonia, low mood, anxiety, trauma, poor concentration, sleep/appetite changes	Depression/anxiety/PTSD	PHQ-9/GAD-7, safety assessment, therapy/medication discussion
Cold intolerance, constipation, weight gain, bradycardia, dry skin	Hypothyroidism	TSH with reflex free T4
Pica, heavy menses, GI bleeding risk, restless legs	Iron deficiency anemia	CBC, ferritin/iron studies, source evaluation
Fever, night sweats, weight loss, lymphadenopathy	Infection, malignancy, inflammatory disease	Directed labs/imaging; avoid reassurance-only approach
Proximal muscle weakness, dark urine, statin or inflammatory symptoms	Myopathy	CK, medication review, neuro/MSK exam
Polyuria, polydipsia, blurry vision, recurrent infections	Diabetes/hyperglycemia	A1c/glucose
Multiple sedating medications, anticholinergics, alcohol/cannabis use	Medication/substance effect	Medication reconciliation and deprescribing plan

Insomnia

Chronic insomnia is difficulty initiating sleep, maintaining sleep, or waking too early despite adequate opportunity for sleep, with daytime impairment. The most important outpatient distinction is insomnia disorder versus insomnia secondary to sleep apnea, restless legs, circadian rhythm disorder, depression/anxiety, substance use, pain, medications, or poor sleep opportunity. Cognitive behavioral therapy for insomnia is first-line for chronic insomnia because it provides durable benefit without medication harms. Sleep hygiene alone is rarely sufficient when insomnia is chronic, but it is useful as part of a structured program.

Insomnia step	Clinical content
1. Define the complaint	Sleep onset, awakenings, early awakening, total sleep time, schedule, naps, shift work, screen/caffeine/alcohol timing, bed partner report
2. Screen for mimics	OSA, restless legs, depression/anxiety, pain, nocturia, GERD, asthma/COPD, medication/substance effects
3. First-line management	CBT-I: stimulus control, sleep restriction, cognitive restructuring, relaxation, consistent wake time
4. Medication role	Short-term adjunct only when needed; avoid chronic sedative dependence, falls, cognitive impairment, and next-day impairment
5. Geriatric caution	Avoid benzodiazepines and Z-drugs when possible due to delirium, falls, fractures, and cognitive adverse effects

Obstructive sleep apnea

OSA causes recurrent upper-airway obstruction during sleep leading to intermittent hypoxemia, arousals, sympathetic activation, and poor sleep quality. Symptoms include loud snoring, witnessed apneas, choking/gasping, morning headaches, dry mouth, nocturia, daytime sleepiness, impaired concentration, and resistant hypertension. Risk increases with obesity, male sex, older age, large neck circumference, craniofacial narrowing, alcohol/sedatives, and hypothyroidism/acromegaly. Untreated OSA contributes to hypertension, atrial fibrillation, pulmonary hypertension, motor vehicle accidents, and impaired quality of life.

STOP-BANG item	Meaning
S - Snoring	Loud snoring
T - Tired	Daytime tiredness/sleepiness
O - Observed	Observed apnea or gasping
P - Pressure	Hypertension
B - BMI	BMI >35 kg/m ²
A - Age	Age >50
N - Neck	Neck circumference >40 cm
G - Gender	Male sex

Treatment centers on weight reduction when appropriate, avoidance of alcohol/sedatives near bedtime, positional therapy for selected patients, continuous positive airway pressure for most clinically significant disease, oral appliances in selected mild-to-moderate cases, and management of nasal obstruction or anatomic contributors. In patients with severe cardiopulmonary disease, opioid use, neuromuscular disease, hypoventilation, or concern for central sleep apnea, in-lab polysomnography is preferred over simple home testing.

8. Outpatient musculoskeletal pain bridge: back, shoulder, knee, and overuse syndromes

Disease-specific rheumatology and orthopedics are covered elsewhere, but ambulatory clinicians frequently evaluate regional pain syndromes. The outpatient task is to distinguish benign mechanical pain from fracture, infection, malignancy, inflammatory arthritis, neurologic compromise, vascular disease, and referred visceral pain. Most acute low back pain without red flags improves with activity, NSAIDs/acetaminophen when appropriate, heat, and reassurance. Imaging in the first several weeks is usually low yield unless red flags are present.

Pain complaint	Benign/common pattern	Red flags	Initial outpatient strategy
Low back pain	Mechanical pain after strain, worse with movement, no neurologic deficits	Cancer history, fever, IV drug use, immunosuppression, trauma, steroid use, saddle anesthesia, bowel/bladder dysfunction, progressive neuro deficit	Stay active, NSAID/acetaminophen if safe, PT if persistent; urgent imaging/referral for red flags
Neck pain	Muscle strain, posture-related, no neuro findings	Trauma, myelopathy, infection, cancer, severe headache, neurologic deficit	Conservative care if uncomplicated; neuro exam and imaging if red flags
Shoulder pain	Rotator cuff tendinopathy, impingement, adhesive capsulitis, biceps tendonitis	Trauma with deformity, infection, cancer, acute neurologic deficit, referred cardiac symptoms	ROM testing, strength testing, PT/NSAIDs, injection in selected cases; image if trauma or persistent weakness
Knee pain	OA, patellofemoral pain, meniscal symptoms, overuse	Hot swollen joint, inability to bear weight, trauma, fever, suspected septic arthritis, DVT signs	Examine effusion/ligaments/meniscus; x-ray based on trauma/age criteria; aspiration if acute hot effusion
Foot/ankle pain	Sprain, plantar fasciitis, tendinopathy	Neurovascular compromise, fracture inability to bear weight, diabetic foot infection, acute ischemia	RICE/brace/PT; imaging using Ottawa rules when applicable

9. Medication-overuse, symptomatic treatment traps, and follow-up safety

Ambulatory treatment frequently fails when symptom control is separated from safety planning. Common traps include repeated antibiotics for viral respiratory illness, chronic PPI use without reassessment, benzodiazepines or Z-drugs for chronic insomnia in older adults, opioids for nonspecific back pain, repeated steroid bursts for undiagnosed inflammatory symptoms, topical steroid overuse on the face or intertriginous areas, and repeated treatment of presumed UTI without culture or alternative diagnosis. A safe plan includes expected time to improvement, what to do if symptoms worsen, when to seek emergency care, and when to return for reassessment.

Clinical trap	Why it matters	Better approach
Antibiotics for acute bronchitis	No benefit for viral illness; adverse effects and resistance	Explain cough duration, assess pneumonia signs, give return precautions
Treating red eye as conjunctivitis without visual acuity/pain assessment	Missed keratitis, uveitis, glaucoma, scleritis	Document visual acuity and screen for pain, photophobia, contact lens use, vision loss
Repeated empiric UTI therapy	Misses STI/vaginitis, stones, malignancy, resistant organisms	Culture atypical/recurrent cases; evaluate vaginal/pelvic symptoms and hematuria
Long-term sedative-hypnotics for insomnia	Falls, delirium, cognitive impairment, dependence	CBT-I first-line; short-term meds only after risk discussion
Imaging uncomplicated acute low back pain	Incidental findings and unnecessary procedures	Reserve imaging for red flags or persistent/progressive symptoms
Treating fatigue with broad panels repeatedly	False positives and missed sleep/mood/medication causes	Focused baseline labs plus structured follow-up and functional assessment

10. Integrated outpatient algorithms

Algorithm: acute cough in clinic

1. Assess ABCs, vital signs, oxygen saturation, work of breathing, mental status, and comorbid risk. If hypoxemic, toxic, hypotensive, confused, or severely dyspneic, send for urgent evaluation.
2. If focal lung exam, fever, tachycardia, tachypnea, pleuritic pain, rigors, older frailty, or clinical concern exists, obtain chest radiograph and evaluate for pneumonia.
3. If no pneumonia features and URI symptoms predominate, diagnose viral URI/acute bronchitis; do not prescribe antibiotics routinely.
4. If wheeze or known obstructive lung disease, consider asthma/COPD exacerbation and bronchodilator/steroid strategy as appropriate.
5. If cough persists beyond 3-8 weeks or has hemoptysis, weight loss, abnormal exam, smoking risk, ACE inhibitor use, GERD/postnasal drip/asthma clues, or immunocompromise, begin chronic cough workup.

Algorithm: red eye in clinic

1. Check visual acuity, pain severity, photophobia, contact lens use, trauma/chemical exposure, vesicular rash, pupil shape/reactivity, and corneal opacity.
2. If pain, photophobia, decreased vision, abnormal pupil, corneal opacity/staining, contact lens risk, trauma, or headache/nausea/halos is present, arrange urgent ophthalmology or ED evaluation.
3. If itching is dominant and bilateral with watery/stringy discharge, treat allergic conjunctivitis.
4. If watery discharge and URI exposure with preserved vision and minimal pain, treat viral conjunctivitis supportively and counsel hygiene.
5. If purulent discharge without red flags, consider topical antibiotics and follow-up if no improvement.

Algorithm: dysuria in clinic

1. Ask about fever, flank pain, vomiting, pregnancy possibility, male sex, immunocompromise, stones, urinary procedures, recurrent resistant organisms, and systemic illness.
2. Ask about vaginal discharge, irritation, pelvic pain, sexual exposure, lesions, or urethral discharge. If present, evaluate for vaginitis/STI rather than treating simple cystitis alone.
3. If classic uncomplicated cystitis pattern and no red flags, treat with a recommended short-course regimen and provide return precautions.
4. If pyelonephritis features are present, obtain urine culture and determine whether outpatient oral therapy is safe. ED/inpatient care is needed for sepsis, pregnancy, vomiting, obstruction, frailty, or immunocompromise.
5. If hematuria persists after infection treatment or gross hematuria occurs, pursue hematuria evaluation rather than repeated empiric antibiotics.

Algorithm: fatigue in clinic

1. Define duration, severity, functional impairment, sleep quantity/quality, mood, medications/substances, exercise tolerance, weight change, fever/night sweats, pain, and objective weakness.
2. Screen for emergency patterns: chest pain, dyspnea at rest, syncope, GI bleeding, severe depression/suicidality, focal neurologic deficits, or rapidly progressive weakness.
3. Perform targeted exam: vitals, BMI, thyroid, lymph nodes, cardiopulmonary, abdominal, neurologic strength, skin/pallor, and edema.
4. Initial labs often include CBC, CMP, TSH, and targeted tests such as ferritin, A1c, pregnancy test, ESR/CRP, HIV/hepatitis, B12, CK, or UA based on history.
5. If initial evaluation is unrevealing, schedule follow-up with sleep, mood, medication, activity, and symptom diary focus rather than leaving the patient without a plan.

11. High-yield ambulatory red flags

Symptom	Red flags
Cough/dyspnea	Hypoxemia, tachypnea, hemoptysis, chest pain, syncope, cyanosis, altered mental status, fever with toxicity, focal lung findings, immunocompromise
Sore throat	Drizzling, stridor, trismus, muffled voice, uvular deviation, neck swelling, inability to swallow secretions, toxic appearance
Sinus/URI	Orbital swelling, proptosis, ophthalmoplegia, vision change, severe headache, meningeal signs, immunocompromise
Red eye	Pain, photophobia, decreased vision, contact lenses, trauma/chemical exposure, corneal opacity, abnormal pupil, halos/nausea
Abdominal/GI	Peritonitis, GI bleeding, persistent vomiting, dehydration, weight loss, anemia, dysphagia/odynophagia, fever, severe focal pain
Dysuria/GU	Fever, flank pain, vomiting, pregnancy, male sex with systemic symptoms, sepsis, obstruction, immunocompromise, gross hematuria
Fatigue	Weight loss, night sweats, fever, lymphadenopathy, objective weakness, dyspnea/chest pain, GI bleeding, suicidal ideation
Back/joint pain	Cancer/infection risk, fever, neurologic deficit, bowel/bladder dysfunction, saddle anesthesia, hot swollen joint, inability to bear weight

12. Quick treatment numbers and thresholds

Topic	Common number / threshold	Clinical use
Acute bacterial sinusitis	Symptoms ≥ 10 days, severe onset, or double worsening	Threshold for considering bacterial sinusitis rather than viral URI
Acute bronchitis	Cough commonly lasts up to several weeks	Counseling prevents unnecessary antibiotics
Uncomplicated cystitis	Nitrofurantoin 100 mg BID x5 days; TMP-SMX DS BID x3 days; fosfomycin 3 g once	Typical short-course regimens when no contraindications
Pyelonephritis	Culture before therapy; reassess within 48-72 h if outpatient	Detect resistance and treatment failure
Chronic diarrhea	>4 weeks	Triggers broader workup for inflammatory, malabsorptive, endocrine, medication causes
Chronic insomnia	≥ 3 nights/week for ≥ 3 months in many definitions	Supports insomnia disorder diagnosis when daytime impairment is present
STOP-BANG	Higher scores increase OSA risk; ≥ 3 often considered increased risk	Screening tool; not diagnostic without sleep testing
Low back pain imaging	Avoid early imaging without red flags, often first 4-6 weeks	Reduces incidental findings and unnecessary procedures
Hoarseness	Persistent $>3-4$ weeks or risk factors	ENT/laryngoscopy evaluation threshold
Fatigue follow-up	Reassess if persistent, progressive, or unexplained after initial evaluation	Prevents both overtesting and abandonment

13. Board-style integration cases

Case stem	Most important next step / diagnosis	Teaching point
Healthy adult with 5 days of cough, colored sputum, normal vitals, clear lungs	Acute bronchitis; no antibiotics	Sputum color does not diagnose bacterial infection

Case stem	Most important next step / diagnosis	Teaching point
Adult with cough, fever, HR 112, O2 sat 91%, focal crackles	Chest radiograph and pneumonia evaluation	Abnormal vitals and focal findings override benign URI assumptions
Sore throat with cough, rhinorrhea, hoarseness	Viral pharyngitis; supportive care	Viral features reduce likelihood of GAS
Red eye in contact lens user with pain and photophobia	Urgent ophthalmology for keratitis/corneal ulcer risk	Contact lens use plus pain is high risk
Dysuria plus vaginal discharge and new partner	STI/vaginitis evaluation	Not all dysuria is cystitis
Fever, flank pain, vomiting, dysuria	Pyelonephritis; culture and assess need for ED/inpatient care	Systemic signs make uncomplicated cystitis unlikely
Older adult with new constipation, weight loss, iron deficiency anemia	Colonoscopy/malignancy evaluation	Alarm features require structural evaluation
Chronic insomnia in older adult requesting sleeping pill	CBT-I first, avoid chronic sedatives	Geriatric adverse effects are high-yield
Snoring, witnessed apneas, resistant HTN, daytime sleepiness	OSA testing	OSA is both a sleep and cardiovascular diagnosis
Fatigue with night sweats, weight loss, lymphadenopathy	Directed malignancy/infection workup	Systemic B symptoms are not simple fatigue

References and source framework

- Uploaded Step-Up ambulatory medicine source packet, pages covering hypertension, hyperlipidemia, and headache, used for chapter continuity and organization.
- CDC adult outpatient treatment recommendations and outpatient antibiotic stewardship principles for acute respiratory infections.
- American College of Physicians clinical practice guideline recommending cognitive behavioral therapy for insomnia as first-line treatment for chronic insomnia disorder in adults.
- Infectious Diseases Society of America guideline summaries for uncomplicated and complicated urinary tract infection treatment principles.
- American Academy of Family Physicians clinical reviews on red eye and painful eye evaluation in primary care.
- Standard internal medicine and primary care frameworks for dyspepsia, GERD, constipation, diarrhea, fatigue, obstructive sleep apnea, and low back pain red flags.